Bowen Sha

PhD Candidate in Computational Chemistry

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Google Scholar · GitHub · LinkedIn · ORCID

Computational chemist specializing in machine-learning force fields, molecular dynamics, and first-principles simulation of energy materials. Passionate about bridging machine learning and quantum chemistry to accelerate materials discovery.

EDUCATION

PhD in Chemistry
Sep 2024 – Present

Massachusetts Institute of Technology · Cambridge, MA, USA

- Thesis: Machine-Learning Force Fields for Single-Atom Catalysis

MSc in Chemistry
Sep 2021 – Jul 2024

Stanford University · Stanford, CA, USA

GPA: 3.9 / 4.0

- Thesis: Reactive Force Fields for Polymer Electrolytes

BSc in Chemistry
Sep 2017 – Jul 2021

University of California, Berkeley · Berkeley, CA, USA

- Minor in Applied Mathematics

RESEARCH EXPERIENCE

Machine-Learning Force Fields for Single-Atom Catalysts
Sep 2024 – Present

Department of Chemistry, Massachusetts Institute of Technology · Cambridge, MA, USA

Developing equivariant graph-neural-network force fields for transition-metal single-atom catalysts.

- Trained a GNN potential on 50k DFT frames, achieving sub-1 meV/atom accuracy
- Reproduced experimentally observed adsorption energies of CO2 on Fe single-atom sites
- Released an open-source training pipeline used by 4 research groups

AI-Driven Molecular Dynamics of Polymer Electrolytes
Sep 2022 – Jul 2024

Department of Chemistry, Stanford University · Stanford, CA, USA

Combined classical and machine-learned force fields to study ion transport in solid polymer electrolytes.

- Elucidated the role of segmental dynamics in lithium-ion conductivity via long-timescale MD
- Predicted a 3x conductivity improvement for a novel copolymer design
- Co-authored 2 papers (J. Phys. Chem. C, ACS Cent. Sci.)

Photocatalytic CO2 Reduction Mechanisms
Jun 2021 – Aug 2022

Lawrence Berkeley National Laboratory · Berkeley, CA, USA

DFT and constrained-CDFT study of CO2 activation on oxide photocatalysts.

- Mapped the free-energy landscape for CO2-to-CO conversion on TiO2 surfaces
- Identified oxygen vacancies as the key active sites, guiding catalyst design
- Contributed to a DOE milestone report on solar fuels

DFT Screening of Metal–Organic Frameworks for Gas Storage Jun 2019 – May 2021

University of California, Berkeley · Berkeley, CA, USA

High-throughput density-functional screening of MOFs for methane and hydrogen storage.

- Screened 1,200+ hypothetical MOFs and shortlisted 12 top candidates
- Correlated pore topology with volumetric uptake, enabling descriptor-based design
- Open-sourced the screening dataset (4.8 GB)

TEACHING EXPERIENCE

Teaching Assistant — Physical Chemistry Jan 2025 – Present

Department of Chemistry, Massachusetts Institute of Technology · Cambridge, MA, USA

- Led discussion sections of 40 students; held weekly office hours
- Authored problem sets and grading rubrics for 5.60 Thermodynamics & Kinetics
- Received a departmental teaching commendation (Spring 2025)

SKILLS

Programming: Python, PyTorch, C++, Shell

Simulation: Gaussian, ORCA, VASP, LAMMPS, GROMACS

Machine Learning: DeepMD-kit, PyTorch Geometric, Scikit-learn, HuggingFace

Tools: Linux, Git, HPC / Slurm, LaTeX

SELECTED PUBLICATIONS

- Sha, B., et al. “Equivariant Graph Neural Network Potentials for Single-Atom Catalysis”, J. Am. Chem. Soc., 2026
- Sha, B., et al. “Ion Transport in Copolymer Electrolytes: A Machine-Learned Force Field Study”, ACS Cent. Sci., 2025
- Sha, B., et al. “Descriptor-Based Screening of Metal–Organic Frameworks”, Chem. Mater., 2024